

Signal Processing for Beyond-5G / 6G — Cheat Sheet

Waveforms • ultra-massive MIMO • RIS • ISAC • AI-native PHY | companion to the 90-min invited talk

1. Sub-THz channel & hardware (100–300 GHz)

Path loss = free-space spreading × molecular absorption:

$$PL(f, d) = \underbrace{\left(\frac{4\pi f d}{c}\right)^2}_{\text{spreading}} \cdot \underbrace{e^{\kappa_{\text{abs}}(f) d}}_{\text{molecular abs.}}$$

O₂/H₂O lines at 60, 119, 183, 325 GHz carve transmission windows; channel sparse, LoS-dominated ⇒ pencil beams.

Rapp PA (AM/AM):

$$g(A) = \frac{GA}{\left[1 + \left(\frac{GA}{A_{\text{sat}}}\right)^{2p}\right]^{1/2p}}$$

Wiener phase noise:

$$\varphi[n+1] = \varphi[n] + w[n], \quad w \sim \mathcal{N}(0, 4\pi^2 f_c^2 c T_s)$$

PA back-off ⇒ PAPR is first-class; phase-noise power $\propto f_c^2$.

2. Waveforms

CP-OFDM symbol:

$$x[n] = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X_k e^{j2\pi k n / N}$$

OFDM ICI power under Doppler spread f_d (symbol T_s):

$$P_{\text{ICI}} \lesssim \frac{(\pi f_d T_s)^2}{3}$$

DFT-s-OFDM: $\mathbf{X} = F_M \mathbf{s}$ (≈ 3 dB lower PAPR).

OTFS — doubly-dispersive input/output:

$$r(t) = \iint h(\tau, \nu) s(t - \tau) e^{j2\pi \nu(t - \tau)} d\tau d\nu$$

ISFFT (DD → TF grid):

$$X_{\text{tf}}[n, m] = \frac{1}{\sqrt{NM}} \sum_{k=0}^{N-1} \sum_{l=0}^{M-1} x[k, l] e^{j2\pi(\frac{nk}{N} - \frac{ml}{M})}$$

DD input-output (2D twisted convolution):

$$y[k, l] \approx \sum_i h_i x[(k - k_i)_N, (l - l_i)_M] e^{j2\pi \phi_i} + w[k, l]$$

SFFT is the inverse of ISFFT; channel is sparse & quasi-static in DD.

AFDM — discrete affine Fourier (chirp) basis:

$$\phi_m[n] = \frac{1}{\sqrt{N}} e^{j2\pi(c_1 n^2 + \frac{m}{N}n + c_2 m^2)}$$

Tune c_1 to the Doppler span; paths separate in the DAFT domain with 1D single-symbol processing.

3. Ultra-massive MIMO & near field

Rayleigh (Fraunhofer) distance, aperture D :

$$d_R = \frac{2D^2}{\lambda} \quad \begin{cases} d > d_R : \text{planar } \mathbf{a}(\theta) \\ d < d_R : \text{spherical } \mathbf{a}(\theta, r) \end{cases}$$

$D=0.5$ m @ 30 GHz ⇒ $d_R=50$ m: users *inside* near field ⇒ range-selective beam focusing.

Hybrid precoding (unit-modulus RF, $|[F_{\text{RF}}]_{ij}|=1/\sqrt{N_t}$):

$$\min_{F_{\text{RF}}, F_{\text{BB}}} \|F_{\text{opt}} - F_{\text{RF}} F_{\text{BB}}\|_F \text{ s.t. } F_{\text{RF}} \in \{\text{array responses}\}$$

Angle-sparse mmWave ⇒ OMP; $N_{\text{RF}} \approx N_s$ chains $\approx$ digital.

CS channel estimation & pilot contamination:

$$\mathbf{y} = \Phi \mathbf{h} + \mathbf{n}, \quad \|\mathbf{h}\|_0 = L \ll \dim, \quad \text{SINR}_j \xrightarrow{M \rightarrow \infty} \frac{\beta_{jj}^2}{\sum_{l \neq j} \beta_{jl}^2}$$

4. Reconfigurable intelligent surfaces

Cascaded channel (N elements):

$$\mathbf{y} = (\mathbf{h}_r^H \Theta \mathbf{G} + h_d^H) \mathbf{x} + n, \quad \Theta = \text{diag}(\beta_1 e^{j\theta_1}, \dots, \beta_N e^{j\theta_N})$$

Optimal phases ⇒ received power scales as N^2 :

$$\theta_n^* = \arg(h_d) - \arg([\mathbf{h}_r]_n^* [\mathbf{G}]_n) \Rightarrow P_{\text{rx}} \propto \left(\sum_n |[\mathbf{h}_r]_n| |[\mathbf{G}]_n| \right)^2 \sim N^2$$

Co-phasing: +6 dB per doubling; random phases ⇒ $P \propto N$. Product path loss $\propto (d_1 d_2)^{-\alpha}$ (passivity tax).

5. Radar & ISAC

Delay / Doppler / resolutions:

$$\tau = \frac{2R}{c}, \quad f_D = \frac{2v f_c}{c}, \quad \Delta R = \frac{c}{2B}, \quad \Delta v = \frac{c}{2f_c T_{\text{obs}}}$$

OFDM radar (RX knows X): $D[n, m] = Y[n, m]/X[n, m]$, then IFFT over n / FFT over m → range-Doppler map. Delay CRB:

$$\text{var}(\hat{\tau}) \geq \frac{1}{8\pi^2 \beta_{\text{rms}}^2 \text{SNR} \cdot N_{\text{obs}}}$$

(β_{rms} = RMS bandwidth ⇒ edge subcarriers are gold.)

6. Multiple access

NOMA (SIC, superposition; near user cancels far):

$$R_{\text{far}} = \log_2 \left(1 + \frac{P_2 |h_f|^2}{P_1 |h_f|^2 + \sigma^2} \right), \\ R_{\text{near}} = \log_2 \left(1 + \frac{P_1 |h_n|^2}{\sigma^2} \right)$$

RSMA (split into common + private):

$$R_k = C_k + R_{p,k}, \quad \sum_k C_k \leq \min_k \log_2(1 + \text{SINR}_{c,k})$$

7. NTN & energy efficiency

LEO Doppler ($v_{\text{sat}} \approx 7.6$ km/s):

$$f_D(t) = \frac{f_c}{c} v_{\text{rel}}(t) \Rightarrow \pm 25 \text{ ppm } (\pm 50 \text{ kHz @ } 2 \text{ GHz})$$

Energy efficiency & ADC power:

$$\eta_{EE} = \frac{R}{P_{\text{tot}}} [\text{bit/J}], \quad P_{\text{ADC}} \propto 2^b f_s$$

Waveform comparison at a glance (relative, qualitative)

	CP-OFDM	DFT-s-OFDM	OTFS	AFDM
PAPR	high	low (−3 dB)	high	moderate
Doppler robustness	poor	poor	excellent	excellent
Diversity (doubly-disp.)	low	low	full	full
Rx / detection cost	low (1-tap)	low + FDE	high (2D MP)	moderate (1D)
Processing domain	TF	TF (SC)	delay-Doppler	DAFT (chirp)
Pilot / CE overhead	moderate	moderate	low (DD impulse)	low
Maturity / standards	deployed (4G/5G)	5G UL	research	emerging (2023+)

No single waveform wins every column ⇒ expect scenario-adaptive PHY. Three formulas organize half the field: $P \propto N^2$ • $d_R = 2D^2/\lambda$ • $\text{var}(\hat{\tau}) \geq 1/(8\pi^2 \beta^2 \text{SNR})$.

8. AI-native physical layer

Model deficit (no faithful model, e.g. PA nonlinearity) vs algorithm deficit (optimum intractable) ⇒ learn the model / the algorithm. **Deep**

unfolding turns K iterations into a K -layer net:

$$\text{ISTA: } \mathbf{x}^{(k+1)} = \eta_{\lambda} (\mathbf{x}^{(k)} + A^H (\mathbf{y} - A \mathbf{x}^{(k)})) \\ \text{LISTA: } \mathbf{x}^{(k+1)} = \eta_{\lambda_k} (W_1^{(k)} \mathbf{y} + W_2^{(k)} \mathbf{x}^{(k)})$$

Learned CSI estimation $\approx$ MMSE *without* knowing statistics; neural

receivers run inside 3GPP frames; end-to-end autoencoder PHY learns the constellation. 3GPP Rel-18/19: CSI-feedback compression, beam prediction, positioning. Open: data, generalization, trust.

Key numbers to remember

- 500 km/h @ 28 GHz $\Rightarrow f_d \approx 13$ kHz ($\varepsilon \approx 0.11$)
- $D=0.5$ m @ 30 GHz $\Rightarrow d_R=50$ m (near field)
- $B=30$ MHz $\Rightarrow \Delta R \approx 5$ m
- RIS N^2 : +6 dB per doubling of N
- LEO 600 km @ 2 GHz $\Rightarrow \pm 50$ kHz swing
- DFT-s-OFDM ≈ 3 dB less PAPR than CP-OFDM

- 3GPP: Rel-19 (ISAC model, AI/ML) $\rightarrow$ Rel-20/21 (6G) $\rightarrow$ spec ~ 2029

Five live MATLAB demos

1. OFDM vs OTFS @ 500 km/h: OFDM BER floor, OTFS keeps falling
2. 64-el hybrid BF: 4 RF chains $\approx$ fully digital
3. RIS $N=16-1024$: slopes 2 (opt.) vs 1 (random)
4. OFDM-ISAC: two moving targets on a range-Doppler map
5. LS / MMSE / NN channel est.: learned $\approx$ MMSE

Reading list — 15 landmark references & surveys

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15. Y. Mao, O. Dizdar, B. Clerckx, R. Schober, P. Popovski, and H. V. Poor, “Rate-splitting multiple access: Fundamentals, survey, and future research trends,” *IEEE Commun. Surveys Tuts.*, vol. 24, no. 4, 2022.

Tools: MATLAB (SPT/CommT/DLT) • NVIDIA Sionna • this talk’s code & Python verification ship together.